

Diff of Trigo. Functions (三角函數微分)

Rule No. 1 - 2:

$$\frac{d}{dx}(\sin x) = \cos x \qquad \frac{d}{dx}(\cos x) = -\sin x$$

e.g. ***1.*** $\frac{d}{dx}(\sin x + \cos x) = \frac{d}{dx}(\sin x) + \frac{d}{dx}(\cos x)$

$= \cos x - \sin x$



2. $\frac{d}{dx}(\sin x - \cos x) =$

3. $\frac{d}{dx}(\sin x \cos x) =$

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Differentiation of Trigonometric Functions

Rule No. 3 - 4:

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

***e.g.* 4.** $\dfrac{d}{dx}(\tan x + \cot x) = \dfrac{d}{dx}(\tan x) + \dfrac{d}{dx}(\cot x)$

$$= \sec^2 x - \csc^2 x$$



5. $\dfrac{d}{dx}(\tan x - 2\cot x) =$

6. $\dfrac{d}{dx}(\sin x \tan x) =$

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Differentiation of Trigonometric Functions

Rule No. 5 - 6:

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

e.g. ***7.*** $\frac{d}{dx}(\sec x + \csc x) = \frac{d}{dx}(\sec x) + \frac{d}{dx}(\csc x)$

$= \sec x \tan x - \csc x \cot x$



8. $\frac{d}{dx}(4\sec x - 3\csc x) =$

9. $\frac{d}{dx}(2\sec x \csc x) =$

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Chain Rule in Differentiation of Trigo. Functions

Differentiation of Composite Trigonometric Functions

Rule No. 7 (Chain Rule – Advance level 鏈式法則進階版) :

$$\frac{d[f(g(x))]^n}{dx} = \frac{d[f(g(x))]^n}{d[f(g(x))]} \cdot \frac{d[f(g(x))]}{d[g(x)]} \cdot \frac{d[g(x)]}{dx}$$



***e.g.* 1.** Differentiate $y = \sin^3 x$ with respect to x.

$$\frac{dy}{dx} = \frac{d(\sin^3 x)}{dx}$$

$$= \frac{d(\sin x)^3}{d(\sin x)} \cdot \frac{d(\sin x)}{dx}$$

$$= 3(\sin x)^2 \cdot (\cos x)$$

$$= 3\sin^2 x \cos x$$

Important Note!!

若三角函數上還有指數時，請首先 **D** 左個指數先!

e.g. 2. Differentiate $y = \cos^4 2x$ with respect to x.

$$\frac{dy}{dx} = \frac{d(\cos^4 2x)}{dx}$$

$$= \frac{d(\cos 2x)^4}{d(\cos 2x)} \cdot \frac{d(\cos 2x)}{d(2x)} \cdot \frac{d(2x)}{dx}$$

$$= 4(\cos 2x)^3 \cdot (-\sin 2x) \cdot (2)$$

$$= -8\sin 2x \cos^3 2x$$

Important Note!!

在這裏用了多次鏈式法則，一次用來 **D** 指數，一次則用來 **D** 三角函數，最後才 **D** 複角。

秘訣：D 完指數 D trigo, 最後再 D Compound Angle !

Chain Rule in Differentiation of Trigo. Functions

[A.Math 2000 Paper I Q.2]

Find

(a) $\dfrac{d}{dx}\sin^2 x$

(b) $\dfrac{d}{dx}\sin^2(3x+1)$ ***(4 marks)***

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HKCEE Questions (Level I)

1. ***[A.Math 2010 Paper I Q.1]***

Find

(a) $\frac{d}{dx}\cos(x^3+1)$,

(b) $\frac{d}{dx}[x\cos(x^3+1)]$. ***(4 marks)***

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HKCEE Questions (Level I)

2. ***[A.Math 1999 Paper I Q.1]***

Find

(a) $\dfrac{d}{dx}\sin(x^2+1)$

(b) $\dfrac{d}{dx}\left[\dfrac{\sin(x^2+1)}{x}\right]$ ***(4 marks)***

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HKCEE Questions (Level I)

3. ***[A.Math 1981 Paper I Q.3]***

If $y = \tan\dfrac{x-1}{x+1}$, find $\dfrac{dy}{dx}$. ***(5 marks)***

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HKCEE Questions (Level I)

4. ***[A.Math 1987 Paper I Q.1]***

Let f(x) = cosec^2 3x. Find f'($\frac{\pi}{12}$). ***(4 marks)***

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HKCEE Questions (Level 2)

5. ***[A.Math 1990 Paper I Q.1]***

Let f(x) = $\sqrt{x^2 + k}\sin 2x$, where k is a constant. If f'(0) = 1, find the value of k.

(5 marks)

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HKCEE Questions (level 2)

6. ***[A.Math 1989 Paper I Q.4]***

Let $\tan\theta = 2\tan x$ and $y = \tan(\theta - x)$ where $0 \le x < \frac{\pi}{2}$.

(a) Express y in terms of tan x.

(b) When $\frac{dy}{dx} = 0$, find the value of x. (Give your answer correct to 3 sig. fig.)

(5 marks)

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HKCEE Questions (Level 2)

7. ***[A.Math 1982 Paper I Q.3]***

Given that $\theta = \dfrac{\sin kt}{3+2\cos kt}$, where k is a non-zero constant,

find the value of $\dfrac{d\theta}{dt}$ when $t = \dfrac{3\pi}{2k}$.

(6 marks)

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